

12.3 Assessment criteria

A: Planning

Ref	Criterion	Mark
P1	Identifies the most appropriate apparatus required for the practical in advance	1
P2	Provides clear details of apparatus required including approximate dimensions and/or component values (for example, dimensions of items such as card or string, value of resistor)	1
P3	Draws an appropriately labelled diagram of the apparatus to be used	1
P4	States how to measure one quantity using the most appropriate instrument	1
P5	Explains the choice of the measuring instrument with reference to the scale of the instrument as appropriate and/or the number of measurements to be taken	1
P6	States how to measure a second quantity using the most appropriate instrument	1
P7	Explains the choice of the second measuring instrument with reference to the scale of the instrument as appropriate and/or the number of measurements to be taken	1
P8	Demonstrates knowledge of correct measuring techniques	1
P9	Identifies and states how to control all other relevant quantities to make it a fair test	1
P10	Comments on whether repeat readings are appropriate for this experiment	1
P11	Comments on all relevant safety aspects of the experiment	1
P12	Discusses how the data collected will be used	1
P13	Identifies the main sources of uncertainty and/or systematic error	1
P14	Plan contains few grammatical or spelling errors	1
P15	Plan is structured using appropriate subheadings	1
P16	Plan is clear on first reading	1
	Maximum marks for this section	16

B: Implementation and measurements

Ref	Criterion	Mark
M1	Records all measurements with appropriate precision, using a table where appropriate	1
M2	Readings show appreciation of uncertainty	1
M3	Uses correct units throughout	1
M4	Refers to initial plan while working and modifies if appropriate	1
M5	Obtains an appropriate number of measurements	1
M6	Obtains measurements over an appropriate range	1
	Maximum marks for this section	6

C: Analysis

Ref	Criterion	Mark
A1	Produces a graph with appropriate axes (including units)	1
A2	Produces a graph using appropriate scales	1
A3	Plots points accurately	1
A4	Draws line of best fit (either a straight line or a smooth curve)	1
A5	Derives relation between two variables or determines constant	1
A6	Processes and displays data appropriately to obtain a straight line where possible, for example, using a log/log graph	1
A7	Determines gradient using large triangle	1
A8	Uses gradient with correct units	1
A9	Uses appropriate number of significant figures throughout	1
A10	Uses relevant physics principles correctly	1
A11	Uses the terms <i>precision</i> and either <i>accuracy</i> or <i>sensitivity</i> appropriately	1
A12	Discusses more than one source of error qualitatively	1
A13	Calculates errors quantitatively	1
A14	Compounds errors correctly	1
A15	Discusses realistic modifications to reduce error/improve experiment	1
A16	States a valid conclusion clearly	1
A17	Discusses final conclusion in relation to original aim of experiment	1
A18	Suggests relevant further work	1
	Maximum marks for this section	18